

BOUNDS FOR CONGRUENCE PRIMES

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Abstract. Given a normalized newform f of weight 2 and level N and rational integral Fourier coefficients, $a_n(f)$, we will be interested in estimating primes ℓ for which there exists a cusp form (not necessarily an eigenform) g of weight 2 and level N with rational integer coefficients $a_n(g)$, such that ℓ divides the difference of all the Fourier coefficients of $a_n(f) - a_n(g)$ with n coprime to N . Such primes are called congruence primes. They are intimately related to the prime divisors of the minimal degree of the modular parametrization of the elliptic curve attached to E via the Eichler-Shimura mapping. We conjecture that this degree is $O(N^{2+\epsilon})$ and show that it is equivalent to the celebrated ABC conjecture. For CM elliptic curves, we prove that this degree is $O(N^{3/2+\epsilon})$. We also study the related question: given two non-isogenous elliptic curves E_1 and E_2 , without complex multiplication and conductors N_1 and N_2 respectively, how large must ℓ be so that the Galois group of $\mathbb{Q}(E_1[\ell], E_2[\ell])$ over $\mathbb{Q}$ is as large as possible.

Dedicated to Goro Shimura

1. Introduction. Let f be a normalized newform of weight 2 for $\Gamma_0(N)$. Suppose that the Fourier coefficients of f are rational integers. We will be interested in those primes ℓ for which there exists a cusp form g (not necessarily an eigenform) of weight 2 and level N , also with integer coefficients such that f and g are orthogonal with respect to the Petersson inner product and

$$f \equiv g \pmod{\ell}.$$

This means that, if $a_n(f)$ and $a_n(g)$ denote the n -th Fourier coefficient of f and g respectively, then

$$a_n(f) \equiv a_n(g) \pmod{\ell}$$

for all n coprime to N . Such a prime ℓ will be called a *congruence prime* for f .

More generally, we are interested in natural numbers δ for which there is a congruence

$$f \equiv g \pmod{\delta}$$

with g orthogonal to f and $a_n(g) \in \mathbb{Z}$. We will then say δ is a *congruence divisor* for f .

Let $\mathcal{S}$ denote the set of cusp forms of weight 2 on $\Gamma_0(N)$ with rational integer Fourier coefficients. Denote by L the orthogonal complement of f in $\mathcal{S}$. Then, one knows that there are only finitely many congruence divisors for f and the largest such, denote it by $D = D_f$, is the exponent of the finite group $\mathcal{S}/(\mathbb{Z}f + L)$ (see Zagier [Z, p. 381]). In [Ma1], Mazur introduces the notion of primes of fusion. That this notion essentially coincides with the concept of a congruence prime has been shown by Ribet [Ri].

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Our aim is to obtain bounds for congruence primes and congruence divisors of f in terms of the level of f . One conjectures that

Conjecture D: $D_f = O(N^{2+\epsilon})$ for any $\epsilon > 0$.

A classical theorem of Eichler and Shimura associates an elliptic curve $E_f/\mathbb{Q}$ such that the L -function of E_f agrees with the L -function of f . More precisely, consider the map

$$\phi(\tau) = 2\pi i \int_{i\infty}^{\tau} f(z) dz \quad (\tau \in \mathcal{H}^*)$$

where $\mathcal{H}^* = \mathcal{H} \cup \mathbb{Q} \cup \{i\infty\}$ and $\mathcal{H}$ denotes the upper half plane. It is easily seen that ϕ is independent of the path chosen from $i\infty$ to τ . If $\gamma \in \Gamma_0(N)$, then

$$\begin{aligned} \omega(\gamma) &:= \phi(\gamma\tau) - \phi(\tau) \\ &= 2\pi i \int_{\tau}^{\gamma\tau} f(z) dz \end{aligned}$$

is independent of τ . Moreover, $\omega : \Gamma_0(N) \rightarrow \mathbb{C}$ is a homomorphism. The image of this map is a lattice of rank 2 in $\mathbb{C}$. If we denote this lattice by Λ_f , then we have the Eichler-Shimura map ϕ_{ES} given by

$$\phi_{ES} : X_0(N) = \mathcal{H}^*/\Gamma_0(N) \rightarrow \mathbb{C}/\Lambda_f.$$

Eichler and Shimura have shown that the torus $\mathbb{C}/\Lambda_f$ is the set of complex points of an elliptic curve E_f defined over $\mathbb{Q}$. An elliptic curve $E/\mathbb{Q}$ which is isogenous to E_f for some f will be called modular.

The map $\phi = \phi_{ES}$ is a branched covering between the two Riemann surfaces $X_0(N)$ and E_f . From the work of Doi, Hida, Ohta, Mazur, Ribet and others, we know that the largest congruence divisor D for f and the topological degree of ϕ are related (see Zagier[Z, p. 381]). The precise relationship has not been spelled out in the literature. For a given modular elliptic curve $E/\mathbb{Q}$, of conductor N , we choose a modular parametrization

$$\phi_E : X_0(N) \rightarrow E$$

of minimal degree. In any $\mathbb{Q}$ isogeny class of modular elliptic curves, there is a curve E with $\deg \phi$ minimal. We call such an E a strong modular elliptic curve. Mazur has shown that if E and E' are isogenous over $\mathbb{Q}$, then there is some isogeny between them of degree ≤ 163 . Thus, $\deg \phi_{E_f} \asymp \deg \phi_E$ for the strong modular elliptic curve in the isogeny class of E_f . We have the following conjecture of Frey [Fr]:

Degree conjecture: For any modular elliptic curve $E/\mathbb{Q}$ of conductor N ,

$$\deg \phi_E = O(N^{2+\epsilon}).$$

The precise relationship between the minimal degree of a modular parametrization of E_f and the congruence divisor $D = D_f$ is given by:

Proposition 0. (Mazur - Ribet) *For a strong modular elliptic curve E with corresponding newform f , we have $\deg \phi_E | D_f$. The only primes dividing $D_f / \deg \phi_E$ are primes dividing the conductor of E .*

Proof. The proof of this theorem will appear in a forthcoming paper by Mazur and Ribet [MR].

Following Cremona [Cr], the topological degree can be computed by observing that

$$\begin{aligned} 4\pi^2(f, f) &= 2\pi^2 i \int_{X_0(N)(\mathbb{C})} f(\tau) d\tau \wedge \overline{f(\tau) d\tau} \\ &= \frac{i}{2} \int_{X_0(N)(\mathbb{C})} (2\pi i f(\tau) d\tau) \wedge \overline{2\pi i f(\tau) d\tau} \\ &= \frac{i}{2} \int_{X_0(N)(\mathbb{C})} \phi^*(dz) \wedge \overline{\phi^*(dz)}, \end{aligned}$$

since $\phi^*(dz) = 2\pi i f(\tau) d\tau$, where $z = x + iy$ is the co-ordinate on E_f . Whence,

$$\begin{aligned} 4\pi^2(f, f) &= \frac{i}{2} (\deg \phi) \int_{E_f(\mathbb{C})} dz \wedge \overline{dz} \\ &= (\deg \phi) \int_{E_f(\mathbb{C})} dx dy \\ &= (\deg \phi) \text{Vol}(E_f) \end{aligned}$$

where $\text{Vol}(E_f)$ is the area of the fundamental parallelogram of the lattice of periods of E_f . As in Cremona [Cr], the Manin constant does not enter into this calculation.

Conjecture D is not unrelated to the celebrated ABC conjecture. In fact, as we shall show below, it is essentially equivalent to it. This relationship is similar in spirit to Szpiro's conjecture [Sz] and its relationship to ABC as exposed in Oesterlé [Oe] or Hindry [H]. In fact, as we shall see below,

$$\log D_f \sim \log N + \frac{1}{6} \log \max\{|c_4|^3, |c_6|^2\}$$

so that Conjecture D is easily seen to be equivalent to Conjecture 4' in [Oe]. The ABC conjecture predicts that if

$$A + B = C \quad \text{with} \quad A, B, C \in \mathbb{Z}$$

and mutually coprime, then

$$\max(|A|, |B|, |C|) \leq \kappa(\epsilon) \left(\prod_{p|ABC} p \right)^{1+\epsilon}$$

for any $\epsilon > 0$ and $\kappa(\epsilon)$ is a constant depending only on ϵ .

To understand the ABC conjecture, Frey [Fr] had the idea of studying the associated elliptic curve

$$E_{A,B,C} : y^2 = x(x - A)(x + B).$$

We will call such curves *Frey curves*. They have conductor

$$N = 2^\delta \prod_{p|ABC} p$$

with δ bounded (see [DK]).

Rubin and Silverberg observed that the modularity of Frey curves follows from Diamond [D] which is an extension of the work of Wiles [W]. Later, Diamond and Kramer [DK] showed that it follows directly from Wiles [W] and Taylor-Wiles [TW]. That is, such a curve is isogenous to a curve of the form E_f for some newform f with $a_n(f) \in \mathbb{Z}$. We have:

Theorem 1.

- (i) *The degree conjecture $\Rightarrow ABC$;*
- (ii) *$ABC \Rightarrow$ the degree conjecture for all Frey curves.*

Remark. A weaker version of (i) was first proved by Frey [Fr]. Namely, he established that some bound for the degree gives some version of ABC (that is, with $1 + \epsilon$ replaced by some larger number). Mai and Murty established (ii) in [MM]. As we shall see below, the precise relationship between the exponents in these two conjectures is determined by invoking a Siegel-type theorem for the lower bound of the symmetric square L -function at $s = 2$ due to Hoffstein and Lockhart [HL].

By Proposition 0, we infer that bounding D_f would give us bounds for $\deg \phi_{E_f}$. In addition, bounding the prime divisors of D_f gives us bounds for the prime divisors of $\deg \phi_{E_f}$. It is this thme that we shall explore in this paper.

For CM elliptic curves, we establish

Theorem 2. *Let $E/\mathbb{Q}$ be a CM elliptic curve with conductor N . Then E is modular and the minimal degree of a modular parametrization satisfies $N^{3/2-\epsilon} \ll \deg \phi_E \ll N^{3/2+\epsilon}$ for any $\epsilon > 0$. (The implied constant may depend on ϵ .)*

For any Frey curve, we have:

Theorem 3. $\log D = O(N^{1/3+\epsilon})$.

Corollary 4. *If ℓ is a congruence prime for f , then*

$$\ell \ll \exp(N^{\frac{1}{3}+\epsilon}).$$

Remark. This result also implies that the number of congruence primes for f is

$$\ll N^{\frac{1}{3}+\epsilon}.$$

The degree can be estimated in several ways. We obtain the following curious properties of various determinants. Let $f_1, \dots, f_r$ be a basis of normalized eigenforms

for $S_2(N)$, the space of cusp forms of weight 2 and level N . Let S be any set of r distinct natural numbers coprime to N . Consider the $r \times r$ matrix

$$A_S = (a_n(f_i))_{\substack{n \in S \\ i \leq r}}.$$

Theorem 5. $(\det A_S)^2$ is a rational integer and

$$D_f^2 | (\det A_S)^2.$$

Using this theorem, we will deduce:

Corollary 6.

$$D_f = O((2N^2r)!^{3/2}(\log 2N^2r)^r)$$

where $r = \dim S_2(N)$.

Mazur [Ma] asked the question: given f as above, let ℓ be a prime such that there is an eigenform g with integer coefficients such that $f \equiv g \pmod{\ell}$. Is ℓ bounded? We will prove:

Theorem 7. Given a newform $f \in S_2(N)$ with $a_n(f) \in \mathbb{Z}$, suppose that there is an eigenform $g \in S_2(N)$ orthogonal to f and with $a_n(g) \in \mathbb{Z}$, and a prime ℓ such that $f \equiv g \pmod{\ell}$. Then $\ell \leq N^{\frac{1}{2}+\epsilon}$.

We indicate in the last section how one can replace $1/2$ by $3/2$ using an easier argument. Recently, Frey has given examples of non-isogenous E and E' over $\mathbb{Q}$, such that $E[11] \simeq E'[11]$ so that $\ell \geq 13$ in Theorem 7.

Theorem 7 has an application to the computation of Galois groups. Recall the well-known theorem of Serre [Ser]. Let E be an elliptic curve $/\mathbb{Q}$ without complex multiplication. Then, the field $\mathbb{Q}(E[\ell])$ obtained by adjoining the ℓ -division points of E has Galois group over $\mathbb{Q}$ isomorphic to $GL_2(\mathbb{F}_\ell)$ for ℓ sufficiently large. How large ℓ should be is answered by means of the effective Chebotarev density theorem. Serre [Ser2] showed that assuming the generalized Riemann hypothesis (GRH) for the Dedekind zeta functions of $\mathbb{Q}(E[\ell])$, one can prove that $\ell \gg (\log N)^2$. It is also possible to show that unconditionally, there is a constant $c > 0$ so that the result holds for $\ell \geq N^c$. (See [Ser3, p. 716].)

Given two distinct non-isogenous elliptic curves E_1 and E_2 over $\mathbb{Q}$, without complex multiplication, how large is

$$\text{Gal}(\mathbb{Q}(E_1[\ell], E_2[\ell])/\mathbb{Q})?$$

The largest it can be is the subgroup

$$H_\ell = \{(a, b) \in GL_2(\mathbb{F}_\ell) \times GL_2(\mathbb{F}_\ell) : \det a = \det b\}$$

of

$$GL_2(\mathbb{F}_\ell) \times GL_2(\mathbb{F}_\ell).$$

Theorem 8. Suppose E_1 and E_2 are non-isogenous modular elliptic curves $/\mathbb{Q}$ without complex multiplication and having conductors N_1 and N_2 respectively. Let $N = \max(N_1, N_2)$. There is an effective constant $c > 0$, such that if $\ell \geq N^c$, then

$$\text{Gal}(\mathbb{Q}(E_1[\ell], E_2[\ell])/\mathbb{Q}) = H_\ell.$$

If we assume the GRH, we can improve this to $\ell \gg \log N$.

Remark. Mazur's conjecture would imply the result for all $\ell \geq c_1$ for some constant c_1 .

2. Proof of Theorem 1.

From the formula already stated:

$$4\pi^2(f, f) = (\deg \phi) \text{Vol}(E_f)$$

we find by Deligne [De, p. 4] and as in Frey [Fr, p. 131] that

$$c^2 4\pi^2(f, f) = (\deg \phi) e^{-2h(E)}$$

where $h(E)$ is the Faltings height of E and c is Manin's constant. By recent work of Edixhoven [E], we know that c is bounded. Therefore

$$2h(E) + \log(f, f) = \log \deg \phi + O(1).$$

By Silverman [Si, p. 258]

$$h(E) = \frac{1}{12}h(j_E) + \frac{1}{12}\log \gamma + O(\log h(j_E)),$$

where $h(j_E)$ is the naive (Weil) height of j_E and γ is the unstable minimal discriminant (see [Si, p. 257]). Writing $j_E = a/b$ with $(a, b) = 1$ and $a, b \in \mathbb{Z}$, we see that the unstable minimal discriminant is Δ/b where Δ is the discriminant of E . It is easily verified that in the case of Frey curves, the unstable minimal discriminant is bounded as the conductor tends to infinity, since

$$\Delta = (ABC)^2/2^8$$

and

$$j_E = \frac{2^8(A^2 + B^2 + AB)^3}{(ABC)^2} > 1.$$

Now $A^2 + B^2 + AB$ is coprime to ABC so that

$$h(j_E) = 3\log(A^2 + B^2 + AB).$$

A simple calculation shows that

$$\log(A^2 + B^2 + AB) = 2\log \max(|A|, |B|, |C|) + O(1).$$

We will now prove that for any $\epsilon > 0$ and N greater than some constant depending only on ϵ , we have the inequality

$$(1 - \epsilon) \log N < \log(f, f) < (1 + \epsilon) \log N.$$

The upper bound is an easy consequence of the Phragmén-Lindelöf theorem applied to the L -function of the symmetric square, as in Mai - Murty [MM]. The L -function attached to the symmetric square of f has the special value

$$L(2, \text{sym}^2(f)) = 288\pi^3 \prod_{p^2|N} \left(1 - \frac{1}{p^2}\right) \frac{(f, f)}{N}.$$

By the Phragmén - Lindelöf theorem, we have

$$L(2, \text{sym}^2(f)) = O(\log N).$$

Hence,

$$(f, f) \leq c_2 N \log N$$

for some constant c_2 . By a result of Hoffstein and Lockhart [HL], we have

$$\log(f, f) > (1 - \epsilon) \log N.$$

Hence

$$\log \max(|A|, |B|, |C|) + O(\log \log \max(|A|, |B|, |C|)) = \log \deg \phi - \log(f, f) + O(1).$$

Therefore, if $\deg \phi \leq N^{2+\epsilon}$, we get the ABC conjecture.

For the converse,

$$\begin{aligned} & \log \max(|A|, |B|, |C|) + O(\log \log \max(|A|, |B|, |C|)) \\ & \geq \log \deg \phi - \log N - \log \log N + O(1). \end{aligned}$$

Therefore, assuming the ABC conjecture gives

$$\deg \phi = O(N^{2+\epsilon}).$$

This proves (ii).

3. Proof of Theorem 2. Recall that if $E/\mathbb{Q}$ has CM, and $j_E \neq 0, 1728$, then E is a quadratic twist of one of a finite set of elliptic curves E_i of conductor N_i and discriminant Δ_i (say). If $j_E = 0$, or 1728 , then it has a Weierstrass equation $y^2 = x^3 + D$ or $y^2 = x^3 + Dx$ respectively (see [Si2, p. 308 and p. 340]). We will discuss the case the curve is a quadratic twist by D (the other cases being handled similarly). In this case, the conductor divides $N_i D^2$ for some i . An easy calculation shows that the discriminant of E is $O(\Delta_i D^6)$. We therefore obtain

$$\log \deg \phi = \log(f, f) + O(1)$$

which implies by the proof of Theorem 1,

$$N_E^{3/2-\epsilon} \ll \deg \phi \ll N_E^{3/2+\epsilon}.$$

This completes the proof of Theorem 2.

4. Proof of Theorem 3.

Stewart and Yu [SY] have proved that for the ABC conjecture, for any $\epsilon > 0$, there is an effectively computable constant $\kappa(\epsilon)$ so that

$$\log \max(|A|, |B|, |C|) \leq \kappa(\epsilon) N^{2/3+\epsilon}.$$

The exponent has recently been improved by Stewart [S] to $1/3 + \epsilon$. Hence, by the above analysis, we deduce that for some constant $\kappa_1(\epsilon)$,

$$\deg \phi < \exp(\kappa_1(\epsilon) N^{1/3+\epsilon}).$$

This completes the proof of Theorem 3.

To deduce Corollary 4, we invoke Proposition 0 to deduce that the primes dividing the degree are precisely the congruence primes apart from primes dividing the level N . Therefore, each prime divisor ℓ of $\deg \phi$ satisfies

$$\ell \leq \exp(\kappa_1(\epsilon) N^{1/3+\epsilon}).$$

5. Proof of Theorem 5.

Let $f_1, \dots, f_r$ be a normalized (i.e. $a_1(f_i) = 1$) orthogonal basis for $S_2(N)$ which are eigenforms for all the Hecke operators T_p for $(p, N) = 1$. (See Lang [L].) Consider $(\det A_S)^2$. Since $A_S A_S^{(t)}$ is invariant under Galois conjugation, we find that $(\det A_S)^2$ is a rational integer. Suppose it is not zero. By going to a finite algebraic extension, we can solve the system

$$\sum_{j \in S} a_j(f_i) m_j = \begin{cases} \det A_S & \text{if } i = 1 \\ 0 & \text{if } i \neq 1 \end{cases},$$

with the m_j algebraic integers. Let g be orthogonal to $f = f_1$ with $a_n(g) \in \mathbb{Z}$. Suppose that

$$f \equiv g \pmod{D}.$$

Let us write

$$g = c_1 f_1 + c_2 f_2 + \dots + c_r f_r,$$

so that $c_1 = 0$. Now consider

$$h(z) = \sum_{j \in S} m_j T_j(g)$$

where T_i denotes the i -th Hecke operator. This is easily seen to be

$$\sum_{i=1}^r c_i \left(\sum_{j \in S} a_j(f_i) m_j \right) f_i$$

which is equal to 0 since $c_1 = 0$. Recall that S consists of numbers coprime to N . Then the n -th Fourier coefficient of $h(z)$ is

$$\begin{aligned} 0 &= \sum_{j \in S} m_j \sum_{d|n, j} a_{nj/d^2}(g) d \\ &\equiv \sum_{j \in S} m_j \sum_{d|n, j} a_{nj/d^2}(f_1) d \pmod{D} \\ &\equiv \sum_{j \in S} m_j a_n(f_1) a_j(f_1) \pmod{D} \\ &\equiv a_n(f_1) \det A_S \pmod{D} \end{aligned}$$

so that choosing $n = 1$ gives that $D^2 | (\det A_S)^2$.

We now prove Corollary 6. By Proposition 2.16 of Shimura [Sh, p. 39], we deduce that the order of vanishing at $i\infty$ of any non-zero cusp form $f \in S_2(N)$ is

$$< \frac{N}{6} \prod_{p|N} \left(1 + \frac{1}{p}\right) = h \quad (\text{say}).$$

Note that $h > 2 \dim S_2(N) = 2r$. Let χ_N be a primitive quadratic character with conductor supported at all the primes dividing N . The r vectors

$$v_i \otimes \chi_N(n) = (a_1(f_i) \chi_N(1), \dots, a_h(f_i) \chi_N(h)) \quad 1 \leq i \leq r$$

corresponding to $f_i \otimes \chi_N \in S_2(M)$ for some $M|N^3$ must be linearly independent for $h = 2 \dim S_2(M)$. Otherwise, we would have a non-zero element in $S_2(M)$ with a zero of order exceeding h contrary to the above remark. Thus, the matrix whose rows are the vectors v_i has rank r . Hence, there are r independent columns. Thus, there is a subset

$$S \subseteq \{1, 2, \dots, h\}$$

consisting of elements coprime to N , so that $|S| = r$ and

$$\det A_S \neq 0.$$

This particular choice allows us to estimate D . Since $h \leq 2 \dim S_2(N^3) = 2N^2r$, we have by Deligne's estimate for the Fourier coefficients:

$$D \leq (2N^2r)!^{3/2} \prod_{i=1}^{2N^2r} d(i)^{1/2}.$$

where $d(i)$ denotes the divisor function. By the arithmetic - geometric mean inequality, we deduce

$$\left\{ \prod_{i=1}^{2N^2r} d(i) \right\}^{1/2N^2r} \leq \frac{1}{2N^2r} \sum_{i=1}^{2N^2r} d(i) \leq \log 2N^2r,$$

by well-known estimates of analytic number theory.

Remark. If N is prime, we need not take S consisting of elements coprime to N since all eigenforms are newforms and thus eigenfunctions for all Hecke operators (including T_p for $p|N$). In this case, we can take S to be

$$\{1, 2, \dots, r\}$$

then we would get the sharper estimate

$$\det A \leq r!^{3/2} \prod_{i=1}^r d(i)^{1/2} \leq r!^{3/2} (\log r)^{r/2}.$$

This is related to the question of whether $i\infty$ is a Weierstrass point for $\Gamma_0(N)$. (Recall that a point x on a compact Riemann surface X is called a Weierstrass point of X if there exists a non-zero differential form whose order of vanishing at x exceeds the genus of X .) If N is a prime, Atkin [A] has shown that $i\infty$ is not a Weierstrass point so that we get the sharper bound for D in this case. It is conjectured that for any squarefree N , $i\infty$ is not a Weierstrass point.

§6. The Rankin-Selberg method.

We collect in this section the necessary facts and estimates derived from the Rankin - Selberg method. All of these facts are contained in the original papers of Rankin and Selberg though they perhaps did not pay much attention to the dependence on the level in the estimation of the error terms. Using standard estimates of the L -function of the symmetric square obtained by invoking the Phragmén - Lindelöf theorem as in [MM], we obtain:

Lemma 9. *Let f be a newform of weight 2 and level N . Then,*

$$\frac{1}{2} \sum_{n \leq x} |a_n(f)|^2 \log^2 \frac{x}{n} = c(f)x^2 + O(x^{3/2}N^{1+\epsilon})$$

where

$$c(f) = \frac{8\pi^3 \phi(N)(f, f)}{N^2}.$$

If g is another newform orthogonal to f , then

$$\frac{1}{2} \sum_{n \leq x} a_n(f) \overline{a_n(g)} \log^2 \frac{x}{n} = O(x^{3/2}N^{1+\epsilon}).$$

Remark. The constant $c(f)$ is given explicitly in Stark [St, p. 90].

If we want estimates for

$$\sum_{\substack{n \leq x \\ (n, M)=1}} |a_n(f)|^2 \log^2 \frac{x}{n}$$

and

$$\sum_{\substack{n \leq x \\ (n, M)=1}} a_n(f) \overline{a_n(g)} \log^2 \frac{x}{n},$$

then this involves removing the Euler factors for primes $p|M$ in the Dirichlet series

$$\sum_{n=1}^{\infty} \frac{|a_n(f)|^2}{n^s}$$

and

$$\sum_{n=1}^{\infty} \frac{a_n(f)\overline{a_n(g)}}{n^s}$$

before applying the method to derive the above asymptotic formulas. This yields

Lemma 10. *For f and g as in Lemma 9,*

$$\frac{1}{2} \sum_{\substack{n \leq x \\ (n, M)=1}} |a_n(f)|^2 \log^2 \frac{x}{n} = c(f, M)x^2 + O(x^{3/2}N^{1+\epsilon}M^\epsilon)$$

and

$$\sum_{\substack{n \leq x \\ (n, M)=1}} a_n(f)\overline{a_n(g)} \log^2 \frac{x}{n} = O(x^{3/2}N^{1+\epsilon}M^\epsilon)$$

where

$$c(f, M) = \frac{8\pi^3 \phi(N)(f, f)}{N^2} \prod_{p|M} \left(\sum_{k=1}^{\infty} \frac{|a_{p^k}(f)|^2}{p^{2k}} \right)^{-1}.$$

We can use this to deduce:

Lemma 11. *If f and g are as in Lemma 9, then there is an $n = O(N^{1+\epsilon}M^\epsilon)$ such that $(n, M) = 1$ and*

$$a_n(f) \neq a_n(g).$$

Proof. We use the estimate of Hoffstein and Lockhart for the Petersson inner product to estimate the residue from below. The result is now immediate.

We briefly indicate the proofs of these lemmas since they do not appear in the literature in the form we need it with the explicit dependence on the level N as stated above. If N is squarefree and $M = 1$, a variant of Lemma 11 is due to Goldfeld and Hoffstein [GH]. A simplified and generalized version of their work is found in [M2].

Let $L(s)$ be a Dirichlet series satisfying the following properties:

- (i) $L(s) = \prod_p L_p(s)$ for $\sigma > 1$ where L_p is the inverse of a polynomial of degree m_p in p^{-s} . The product is over all finite primes.
- (ii) There is a positive integer d so that $m_p \leq d$ for all p and $m_p = d$ for all but finitely many p .

Let us write

$$L_p(s) = \prod_{i=1}^{m_p} (1 - \pi_{i,p} p^{-s})^{-1}$$

where $\pi_{i,p} \in \mathbb{C}$, $|\pi_{i,p}| = 1$. Define

$$L_p^v(s) = \prod_{i=1}^{m_p} (1 - \bar{\pi}_{i,p} p^{-s})^{-1}$$

and set

$$L^v(s) = \prod_p L_p^v(s).$$

Let A be a positive real number and a, b non-negative integers with $a + b \leq d$. Set

$$\Lambda(s) = A^{s/2} (\pi^{-s/2} \Gamma(s/2))^a (\pi^{(s+1)/2} \Gamma((s+1)/2))^b L(s)$$

$$\Lambda^v(s) = A^{-s/2} (\pi^{-s/2} \Gamma(s/2))^a (\pi^{-(1+s)/2} \Gamma((1+s)/2))^b L^v(s).$$

Suppose that

- (iii) $\Lambda(s)$ and $\Lambda^v(s)$ have an analytic continuation to the entire complex plane except possibly for a pole at $s = 0$ or $s = 1$,
- (iv) $\Lambda(s) = w \Lambda^v(1-s)$ with $w \in \mathbb{C}$, $|w| = 1$. Under assumptions (i) to (iv), we have for $0 \leq \sigma \leq 1$,

$$|L(\sigma + it)| \leq (A\tau^d)^{\frac{1-\sigma}{2}} (\log A\tau^d)^d$$

where $\tau = |t| + 2$. This is an effective version of the Phragmén-Lindelöf theorem. The proof given in [MM2] applies to functions with no pole at $s = 0$ or 1 . But it can easily be adapted to give the same result if $L(s)$ has a pole at $s = 0$ or 1 . (Let us also take this opportunity to point out a misprint in [MM2]. On page 251 of [MM2] in condition (i), L_p is the *inverse* of a polynomial in p^{-s} .)

We apply this result to the series

$$L_{f,g}(s) = \zeta_N(2s) \sum_{n=1}^{\infty} \frac{a_n \bar{b}_n}{n^{s+1}}$$

where

$$\zeta_N(2s) = \prod_{p \nmid N} \left(1 - \frac{1}{p^{2s}}\right)^{-1}.$$

Rankin [R] established the analytic continuation of $L_{f,g}(s)$. Ogg [Ogg], derived the functional equation in the case f and g have squarefree levels. Li[Li] has generalized this to arbitrary level and weight.

For the sake of simplicity, we will confine our attention to squarefree level. An easy adaptation of the method to the non-squarefree level is effected by using the (slightly more complicated) functional equation of Li [Li].

Let f, g be normalized cusp forms of squarefree level N_1 and N_2 respectively. Let $N = \text{lcm}(N_1, N_2)$ and $M = \text{gcd}(N_1, N_2)$. For $p|M$, let

$$c_p = a_p b_p = \pm 1$$

and set

$$\Phi^+(s) = \Phi(s) \prod_{p|M} \left(1 - c_p p^{-s}\right)^{-1}$$

where

$$\Phi(s) = \left(\frac{\sqrt{N}}{2\pi}\right)^{2s} \Gamma(s) \Gamma(s+1) L_{f,g}(s).$$

Then

$$\Phi^+(1-s) = \Phi^+(s).$$

We now consider the integral

$$\frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \left(\sum_{n=1}^{\infty} \frac{a_n \bar{b}_n}{n^{s+1}} \right) \frac{x^s}{s^3} ds$$

which is equal to

$$\frac{1}{2} \sum_{n \leq x} \frac{a_n \bar{b}_n}{n} \log^2 \frac{x}{n}.$$

We write the Dirichlet series in the integrand as

$$\frac{1}{\zeta_N(2s)} \left(\zeta_N(2s) \sum_{n=1}^{\infty} \frac{a_n \bar{b}_n}{n^{s+1}} \right),$$

and apply the Phragmén - Lindelöf theorem to the paranthetic term to obtain an estimate of

$$O(N^{1/2} T \log^4 NT)$$

on the line $\Re(s) = 1/2$. The term $1/\zeta_N(2s)$ can be estimated easily on $\Re(s) = 1/2$. We derive the estimate

$$\frac{1}{2} \sum_{n \leq x} \frac{a_n \bar{b}_n}{n} \log^2 \frac{x}{n} = O(x^{1/2} N^{1/2} \log^4 N).$$

when $f = g$, we find

$$\sum_{n \leq x} \frac{|a_n|^2}{n} \log^2 \frac{x}{n} = 2\alpha_f x + O(x^{1/2} N^{1/2} \log^4 N)$$

where

$$\alpha_f = \frac{4\pi^3 \phi(N)(f, f)}{N^2}.$$

Thus,

$$\frac{1}{2} \sum_{n \leq x} (a_n - b_n)^2 \log^2 \frac{x}{n} = (\alpha_f + \alpha_g)x + O(x^{1/2} N^{1/2} \log^4 N).$$

This leads to the result that

$$a_n \neq b_n$$

for some $n = O(N \log^8 N)$. (In [M2], the slightly better estimate $O(N \log \log N)$ is obtained.) This completes the sketch of the proof of Lemmas 9, 10 and 11.

§7. Proof of Theorem 7.

If we have a congruence between newforms f and g

$$a_n(f) \equiv a_n(g) \pmod{\ell}$$

for all $(n, N) = 1$, then by Lemma 11, we deduce that for some $n = O(N^{1+\epsilon} \ell^\epsilon)$, we have

$$a_n(f) \neq a_n(g)$$

and $(n, N) = 1$. Therefore,

$$\ell \ll N^{\frac{1}{2}+\epsilon}.$$

This completes the proof of Theorem 7.

§8. Proof of Theorem 8.

By Serre [Se], we know that if $\ell \geq N^c$ for some constant c , then

$$\text{Gal}(\mathbb{Q}(E_1[\ell])) \simeq GL_2(\mathbb{F}_\ell)$$

and

$$\text{Gal}(\mathbb{Q}(E_2[\ell])) \simeq GL_2(\mathbb{F}_\ell)$$

We now apply results of Ribet [Ri] to deduce that the only obstruction to the assertion of the theorem is the set of congruence primes linking f_1 and f_2 where f_1 and f_2 are the newforms associated to E_1 and E_2 . We will need the following lemma which is a special case of Theorem 3.8 of [Ri2].

Lemma 12. *For a prime $\ell \geq 5$, let e be a divisor of $\ell - 1$, let V, V' be vector spaces of dimension 2 over $\mathbb{F}_\ell$, let B, B' be the subgroups of $GL(V)$ and $GL(V')$ respectively consisting of all elements whose determinants are e -th powers. Let E be the subgroup of $B \times B'$ consisting of all (b, b') with $\det b = \det b'$ and suppose H is a subgroup of E in $B \times B'$ whose projections to each factor are surjective. Then, if $H \neq E$, there is an isomorphism ϕ from V to V' and a character χ of H with $\chi^2 = 1$ such that*

$$b' = \chi(h) \phi b \phi^{-1}$$

for all $(b, b') \in H$.

We apply this lemma in the case $e = 1$. We suppose ℓ is a prime such that $\ell \geq N^c$ with c sufficiently large and $G_i = \text{Gal}(\mathbb{Q}(E_i[\ell])/\mathbb{Q}) \simeq GL_2(\mathbb{F}_\ell)$ for $i = 1, 2$. Then, let $H = \text{Gal}(\mathbb{Q}(E_1[\ell], E_2[\ell])/\mathbb{Q}) \subseteq G_1 \times G_2$ and E the subgroup of elements (b, b') of $G_1 \times G_2$ with $\det b = \det b'$. Clearly H satisfies the conditions of the lemma. If $H \neq E$, there is an isomorphism ϕ and a quadratic character χ of H such that

$$b' = \chi(h) \phi b \phi^{-1} \quad \forall (b, b') \in H.$$

Thus, evaluating this on the powers of the Frobenius automorphism gives

$$a_{p^m}(E_1) \equiv \chi(p^m) a_{p^m}(E_2) \pmod{\ell}$$

for all

$$p \nmid \ell N_1 N_2.$$

Since E_1 and E_2 are modular, there are cusp forms f_1 and f_2 such that

$$a_{p^m}(E_i) = a_{p^m}(f_i), \quad i = 1, 2.$$

Therefore,

$$a_n(f_1) = \chi(n)a_n(f_2) \pmod{\ell}$$

for all n coprime to $\ell N_1 N_2$. Moreover, the quadratic character χ can only be ramified at most at the prime dividing ℓ , since the kernel of χ cuts out a quadratic extension contained in $\mathbb{Q}(E_i[\ell])$ and the quadratic subfield of $\mathbb{Q}(\zeta_\ell)$ is the only such. By Lemma 11, we deduce that for some $n = O(N^{1+\epsilon}\ell^\epsilon)$, and $(n, \ell N_1 N_2) = 1$, we deduce

$$a_n(f_1) \neq \chi(n)a_n(f_2).$$

Therefore, by the Ramanujan estimate, we deduce $\ell \ll N^{1/2+\epsilon}$. The first assertion now follows from Theorem 7.

To deduce the statement on the GRH, we know from Serre that the Galois groups generated by the ℓ -division points of E_1 and E_2 are as large as possible for $\ell \geq (\log N)^2$. By the Rankin-Selberg method, we know that for $i = 1, 2$,

$$\sum_{p \leq x} |a_p(f_i)|^2 \log p = x^2 + O(x(\log Nx)^2)$$

and

$$\sum_{p \leq x} a_p(f_1) \overline{a_p(f_2)} = O(x(\log Nx)^2)$$

from which we deduce that

$$\sum_{p \leq x} \left(a_p(f_1) - a_p(f_2) \right)^2 = 2x^2 + O(x(\log Nx)^2)$$

so that for some $p_0 \ll (\log N)^2$, we have

$$a_{p_0}(f_1) \neq a_{p_0}(f_2).$$

Hence, ℓ divides

$$|a_{p_0}(f_1) - a_{p_0}(f_2)|$$

so that

$$\ell \ll (\log N)$$

which completes the proof of Theorem 8.

9. An alternate method.

We make some remarks using an alternate method. At the outset, let us note that optimal results are obtained by the Rankin-Selberg method as in Theorem 7. Also, assuming the generalized Riemann hypothesis, one can bound the congruence prime ℓ by $O(\log N)$ using the method outlined in the proof of Theorem 8. If

instead of the generalised Riemann hypothesis, we assume only the generalized Lindelöf hypothesis for the Rankin-Selberg convolutions that intervene in the proof of Theorem 7, then the term $N^{1/2+\epsilon}$ appearing in the error terms in Lemmas 9, 10, and 11 can be replaced by N^ϵ . This would then give an estimate of $\ell \ll N^\epsilon$ in Theorem 7.

Without the Rankin-Selberg method, one can prove:

Theorem 7*. *Let f and g be as in Theorem 7 and suppose that*

$$f \equiv g \pmod{\ell}.$$

Then, $\ell \ll N^{3/2+\epsilon}$.

Proof. We may suppose $\ell > N$. We have

$$a_n(f) \equiv a_n(g) \pmod{\ell}$$

for all $(n, N) = 1$. We twist by a quadratic character χ_N whose conductor is divisible by all the primes dividing N . Then,

$$a_n(f)\chi_N(n) \equiv a_n(g)\chi_N(n) \pmod{\ell}$$

for all n . The level of the twist of f by χ_N divides N^3 . Therefore, if

$$a_n(f)\chi_N(n) = a_n(g)\chi_N(n)$$

for all

$$n \leq \sum_{d|N^3} \dim S_2(d) = O(N^{3+\epsilon})$$

then

$$a_n(f) = a_n(g)$$

for all $(n, N) = 1$ contradicting the assumption $(f, g) = 0$. Hence, for some $n = O(N^{3+\epsilon})$, we have

$$a_n(f)\chi_N(n) \neq a_n(g)\chi_N(n).$$

Reasoning now as before, we deduce that

$$\ell \ll N^{3/2+\epsilon}.$$

This argument may be of some independent interest. It is similar to arguments used in [M2].

10. Concluding remarks.

Masser and Wusthölz [MW] have obtained results similar to Theorem 8. Their bounds on how large ℓ should be to ensure that the Galois groups be large as possible are given in terms of the Faltings height of the elliptic curves whereas the results above are in terms of the conductor of N .

REFERENCES

- [A] A.O.L. Atkin, Weierstrass points at cusps of $\Gamma_0(N)$, *Annals of Math.*, (2) **85** (1967) pp. 42-45.
- [Cr] J. Cremona, Computing the degree of the modular parametrization of a modular elliptic curve, *Math. Comp.* (1996)
- [De] P. Deligne, Preuve des conjectures de Tate et de Shafarevitch (d'après G. Faltings), *Sém. Bourbaki*, **616** (1983).
- [D] F. Diamond, On deformation rings and Hecke rings, *Ann. of Math.* (2) **144** (1996), no. 1, p. 137-166.
- [DK] F. Diamond and K. Kramer, Modularity of a family of elliptic curves, *Inter. Math. Res. Notices* (1996).
- [E] B. Edixhoven, Modular parametrizations at primes of bad reduction, in preparation.
- [Fr] G. Frey, Links between elliptic curves and solutions of $A - B = C$, *Journal of the Indian Math. Soc.*, **51** (1987) pp. 117 - 145.
- [GH] D. Goldfeld and J. Hoffstein, On the number of Fourier coefficients that determine a modular form, *Contemp. Math.*, **143** (1993) p. 385 - 393.
- [H] M. Hindry, "a,b,c", conducteur, discriminant, Publications mathématiques de l'université Pierre et Marie Curie, 84, 1986-87.
- [HL] J. Hoffstein and P. Lockhart, Coefficients of Maass forms and the Siegel zero, *Annals of Math.*, (2) **140** (1994), no. 1, pp. 161 - 181.
- [La] S. Lang, Introduction to modular forms, Springer-Verlag, Berlin, 1995.
- [Li] W. Winnie-Li, L-series of Rankin type and their functional equations, *Math. Annalen*, **244** (1973) p. 135 - 166.
- [Ma] B. Mazur, Questions about number, to appear.
- [Ma1] B. Mazur, Modular curves and the Eisenstein ideal, *Inst. Hautes Études Sci. Publ. Math.*, **47** (1977) pp. 33 - 186.
- [Ma2] B. Mazur, Rational isogenies of prime degree, *Inventiones Math.*, **44** (1978), no.2, 129-162.
- [MaR] B. Mazur and K. Ribet, Degrees of modular parametrizations and congruences between modular forms, to appear.
- [MM] L. Mai and R. Murty, The Phragmén - Lindelöf Theorem and Modular Elliptic Curves, *Contemporary Math.*, **166** (1994) pp. 335 - 340.
- [MM2] M. Ram Murty and V. Kumar Murty, A variant of the Bombieri - Vinogradov theorem, in Number Theory (edited by H. Kisilevsky and J. Labute).
- [M2] R. Murty, Congruences between modular forms, in Analytic Number Theory, London Mathematical Society Lecture Note Series, **247** (edited by Y. Motohashi), 1997, pp. 309-320.
- [MW] D.W. Masser and G. Wüstholz, Galois Properties of Division Fields of Elliptic Curves, *Bull. London Math. Soc.*, **25** (1993) pp. 247 - 254.

- [Oe] J. Oesterlé, Nouvelle approches du “Théorème” du Fermat, Sémin. Bourbaki, 1987-88, exposé 694, Astérisque 161- 162 (1988).
- [Ogg] A. P. Ogg, On a convolution of L -series, *Inv. Math.*, **7** (1969), pp. 297 - 312.
- [R] R. Rankin, Contributions to the theory of the Ramanujan’s τ -function and similar arithmetical functions, I, II, *Proc. Cambridge Phil. Soc.*, **35** (1939) pp. 351-372.
- [Ri] K. Ribet, Mod p Hecke operators and congruences between modular forms, *Inv. Math.*, **71** (1983), 193 - 205.
- [Ri2] K. Ribet, In ℓ -adic representations attached to modular forms, *Inventiones Math.*, **28** (1975) 245-275.
- [RS] K. Rubin and A. Silverberg, On Wiles’ Cambridge lectures, *Bull. Amer. Math. Soc.*, **31** (1994), 15-38.
- [Ser] J.-P. Serre, Propriétés galoisienne des points d’ordre fini des courbes elliptiques, *Inventiones Math.*, **15** (1972), 259 - 331.
- [Ser2] J.-P. Serre, Quelques applications du théorème de densité de Chebotarev, *Publ. Math. IHES*, **54** (1981), 123 - 301. (see also, Volume 3 of Collected Papers, Springer - Verlag).
- [Ser3] J.-P. Serre, Collected Papers, Volume 3, Springer - Verlag.
- [Sh] G. Shimura, Introduction to the arithmetic theory of automorphic functions, *Publ. Math. Soc. Japan* **11** (1971).
- [Sil] J.H. Silverman, Heights and Elliptic curves, in Arithmetic Geometry, edited by G. Cornell and J.H. Silverman, Springer-Verlag, 1986, p. 253-265.
- [Si2] J.H. Silverman, The Arithmetic of Elliptic Curves, Springer-Verlag, 1986.
- [Sh] G. Shimura, Introduction to the arithmetic theory of automorphic functions, *Publ. Math. Soc. Japan*, No. 11, 1971.
- [St] H.M. Stark, L-functions at $s = 1$, II, Artin L-functions with Rational Characters, *Advances in Math.*, **17** (1975) p. 60-92.
- [S] C. Stewart and K. Yu, On the abc conjecture, II, to appear.
- [SY] C.L. Stewart and K. Yu, On the abc conjecture, *Math. Annalen*, **291** (1991) pp. 225 - 230.
- [Sz] L. Szpiro, La conjecture de Mordell (d’après Faltings), Sémin. Bourbaki, 1983-84, exposé 619, Astérisque 121-122 (1985), 93-103.
- [TW] R. Taylor and A. Wiles, Ring theoretic properties of certain Hecke algebras, *Annals of Math.*, **141** (1995), 553 - 572.
- [W] A. Wiles, Modular elliptic curves and Fermat’s last theorem, *Annals of Math.*, **141** (1995), 443 - 551.
- [Z] D. Zagier, Modular parametrization of elliptic curves, *Can. Math. Bulletin*, **28** (1985) 372 - 384.

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